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6.9 Answer Explanations: Two-Part Analysis

The following discussion of Data Insights is intended to familiarize you with the most efficient and effective approaches to the kinds of problems common to Data Insights. The particular questions in this chapter are generally representative of the kinds of Data Insights questions you will encounter on the GMAT exam. Remember that it is the problem-solving strategy that is important, not the specific details of a particular question.

Questions 114 to 123 — Difficulty: Easy

114. Two employees of Company X, Winnie and Ali, have been informed that they will each receive bonuses this year, and that, for these bonuses, they will share a total amount of 560,000 KES (Kenyan shillings). Winnie makes a base salary of 4,600,000 KES per annum, and Ali makes a base salary of 3,400,000 KES per annum.

    Assuming that the two employees receive the same percentage of their respective per annum base salaries as bonuses, indicate below the bonus amount that Winnie will receive and the bonus amount that Ali will receive. Make only two selections, one in each column.

Winnie	Ali	
		216,000
		238,000
		256,000
		304,000
		322,000

Answer Explanation:

Infer

***Winnie*:**

Winnie and Ali will each get the same percentage of their per annum base salaries as bonuses. That means that Winnie's bonus must be the same percentage of her and Ali's combined total bonus amount of 560,000 KES that her per annum base salary is of their combined total per annum base salaries of 4,600,000 KES + 3,400,000 KES = 8,000,000 KES. Thus, Winnie will get a bonus of W KES such that = $\frac{W}{560{,}000} = \frac{4{,}600{,}000}{8{,}000{,}000} = 0.575$.

Therefore, Winnie's bonus is $W = 0.575 \times 560{,}000 = 322{,}000$.

The correct answer is *322,000*.

***Ali*:**

We saw above that Winnie will get 322,000 KES of the total 560,000 KES bonus she's splitting with Ali. Thus, Ali's share of the bonus will be 560,000 KES − 322,000 KES = 238,000 KES.

The correct answer is *238,000*.

115. After a 16-member team at Company A saw a certain software demonstration, each of the team members gave the software exactly one of the ratings 1, 2, 3, 4, or 5, where greater numbers indicate higher ratings. The team's manager intended to record the 16 ratings but accidentally recorded one team member's rating twice. The ratings she recorded show two ratings of 1, two ratings of 2, two ratings of 3, two ratings of 4, and nine ratings of 5, for a total of 17 ratings.

    The manager determined that the extra rating was ____P____, and after this extra rating was removed, the average (arithmetic mean) of the remaining 16 ratings was exactly ____Q____.

    Select for P and for Q the options that create a statement that is consistent with the information provided. Make

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Select for P and for Q the options that create a statement that is consistent with the information provided. Make only two selections, one in each column.

P	Q	
		1
		2
		3
		4
		5

Answer Explanation:

Infer

***P*:**

All the response options are whole numbers. One of them must equal Q, the arithmetic mean of the 16 ratings other than P (i.e., the extra rating). Since Q is a whole number, the sum of those 16 ratings must be divisible by 16. That is, the sum of all 17 ratings minus P is divisible by 16. The 17 ratings sum to 2 × (1 + 2 + 3 + 4) + 9 × 5 = 65. And 65 − 1 = 64 = 4 × 16. Since the sum of the 17 ratings minus 1 is divisible by 16, $P = 1$.

The correct answer is *1*.

***Q*:**

The mean of the 16 ratings other than P is the sum of those ratings divided by 16. We saw above that the sum of those 16 ratings is 64, so their mean is 64/16 = 4.

The correct answer is *4*.

116. Natural gas, when burned as fuel for heat, produces about 1,000 BTU of heat per cubic foot of gas when fully burned. Propane, another gas used as fuel, produces about 2,500 BTU per cubic foot of gas when fully burned.

Furnaces burning either fuel burn at various efficiencies, according to the proportion of heat energy available from the fuel that actually contributes to the heating of the building to be heated (as opposed, e.g., to escaping up a chimney in the form of dissipated heat or unburned fuel).

Kgomotso is shopping for a furnace that will produce about 50,000 BTU per hour of usable heat. None of the available furnaces burns both propane and natural gas. Also, no two of them operate at the same efficiency. In the table, identify, if possible, a price per cubic foot of *Propane* and a price per cubic foot of *Natural gas*, such that the fuel-cost per BTU of usable heat is the same using either fuel. If this determination is impossible with the given information, then choose "Undeterminable" in each column. Make only two selections, one in each column.

Propane	Natural gas	Price per cubic foot
		$0.0035
		$0.0070
		$0.0110
		$0.0175
		$0.0350

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		Undeterminable

Answer Explanation:

Infer

Let $p\%$ and $n\%$ be the percent of heat energy available to heat a building (i.e., usable heat) for a furnace that burns propane and a furnace that burns natural gas, respectively. If P and N represent the price per cubic foot of propane and natural gas, respectively, then $\left(\frac{p}{100}\right)(2{,}500)(\mathrm{P})$ represents the price per BTU of usable heat using propane and $\left(\frac{n}{100}\right) - (1{,}000)(\mathrm{N})$ represents the price per BTU of usable heat using natural gas. The task is to determine, if possible, P and N such that $\left(\frac{p}{100}\right)(2{,}500)(\mathrm{P}) = \left(\frac{n}{100}\right)(1{,}000)(\mathrm{N})$ or equivalently, $5pP = 2nN$.

***Propane*:**

Since p and n vary from furnace to furnace and values are not included in the given information, it is impossible to determine P, the price per cubic foot of propane for which the fuel-cost per BTU of usable heat will be the same as that for natural gas.

The correct answer is *Undeterminable*.

***Natural Gas*:**

Since p and n vary from furnace to furnace and values are not included in the given information, it is impossible to determine N, the price per cubic foot of natural gas for which the fuel-cost per BTU of usable heat will be the same as that for propane.

The correct answer is *Undeterminable*.

117. At Company X, a new firm which is at present devoted entirely to the design of software applications, all employees are required to agree to the following policy:

Any idea or invention conceived of by employees that is related to Company X's past, present, or future business is owned exclusively by Company X, not the individual employee.

Thus, if a Company X employee has conceived of an idea or invention during his/her employment with Company X related in any manner to Company X's business, that invention is owned by Company X. The employee may not use or exploit that idea or invention for his or her personal benefit.

An employee of Company X has recently sold a children's novel to a publisher. Company X has claimed ownership of the novel on the basis of the stated policy, and the employee intends to dispute the company's claim.

If ___1___, then a successful defense of the company's claimed ownership rights would likely include a statement that ___2___.

Select for *1* and for *2* the options whereby the statement is most clearly supported. Make only two selections, one in each column.

1	2	
		mere use of Company X's resources in an activity is sufficient grounds for regarding that activity as related to Company X's business
		mere use of Company X's resources in an activity is not sufficient grounds for regarding that activity as related to Company X's business
		the employee wrote the novel on a computer owned by the company
		the employee has received a significant amount of money from the sale

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		the company has plans to expand into publishing

Answer Explanation:

Strategize

Note that because the two blanks are part of the same conditional sentence, you must consider both responses together.

***1* and *2*:**

All employees are required to agree to a policy stating that any idea or invention conceived by employees that is related to the business of Company X belongs to that company. On the basis of this policy, the company claims ownership of the children's novel that a Company X employee has sold. A successful defense of the company's ownership rights would link the novel as an idea or invention in some way to the company's business. Suppose it is the case that the employee used a computer owned by Company X. Then a successful defense of the claimed ownership right would likely appeal to a claim that the mere use of Company X's resources (such as a computer) in an activity is sufficient grounds for regarding that activity as related to Company X's business.

The correct answer to 1 is *the employee wrote the novel on a computer owned by the company.*

The correct answer to 2 is *mere use of Company X's resources in an activity is sufficient grounds for regarding that activity as related to Company X's business.*

118. A village's ordinances require residents to exercise reasonable care to avoid disturbing their neighbors with noise. The village council may impose fines for noise violations in response to formal complaints. But unless there have been formal complaints from multiple residents, the council will not consider a complaint if the complaining individual has not attempted to resolve the issue directly with the alleged violator or with the violator's landlord if the violator is a renter. The council's first action in response to any formal verbal or written complaint that they are considering will be to offer to mediate the dispute.

Select for *1* and for *2* two different events such that the guidelines most clearly indicate that if the event selected for *1* occurs, then the event selected for *2* either will occur or has already occurred. Make only two selections, one in each column.

1	2	
		The council imposes a fine for noise violation.
		The council considers a formal written noise complaint.
		The complaining individual attempts to resolve the issue with the alleged violator.
		The council offers to mediate a dispute about noise.
		The complaining individual contacts the alleged violator's landlord.

Answer Explanation:

Recognize

The final sentence implies that, in response to any formal verbal or written complaint made to the council, the council will offer to mediate the dispute. Therefore, if the council is considering a formal written noise complaint, then the council will offer to mediate the dispute about noise.

***1*:**

The correct answer is *The council considers a formal written noise complaint.*

***2*:**

The correct answer is *The council offers to mediate a dispute about noise.*

119. Manager S: Our company provides a small annual monetary benefit for each employee who submits

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119. Manager S: Our company provides a small annual monetary benefit for each employee who submits documentation to the company proving that the employee had a comprehensive medical exam in a given year. The policy has been successful in increasing the number of employees who get the exam annually precisely because the benefit requires this specific form of documentation, and this success has resulted in improved employee health. **[Insert Sentence 1.]**

Manager T: I agree that the overall policy has improved employee health, and it is good to improve employee health. However, the required documentation resulted in our company having access to an employee's private medical data. The benefit to employee health is not worth compromising employee privacy. **[Insert Sentence 2.]**

Select for *Sentence 1* the sentence that best completes Manager S's argument, and select for *Sentence 2* the sentence that best completes Manager T's argument. Make only two selections, one in each column.

Sentence 1	**Sentence 2**	
		Requiring the documentation has no effect on employees.
		Our company should continue requiring the specific documentation that it does.
		Our company should not require that form of documentation.
		Promoting employee health with financial incentives is counterproductive.
		Promoting employee health with financial incentives is worthwhile.

Answer Explanation:

Evaluate

***Sentence 1*:**

Each of the first, third, and fourth answer choices at least somewhat conflicts with Manager S's views. Also, although the fifth answer choice is in agreement with Manager S's views, it does not mention documentation, which is an important aspect of Manager S's argument. Therefore, only the second answer choice could support Manager S's argument. Moreover, since Manager S argues that the specific form of documentation now used is desirable, a concluding sentence (such as the second answer choice) that the company should continue requiring the specific documentation that it does would complete Manager S's argument.

The correct answer is *Our company should continue requiring the specific documentation that it does*.

***Sentence 2*:**

Each of the first, second, and fourth answer choices at least somewhat conflicts with Manager T's views. Also, although the fifth answer choice is in agreement with Manager T's views, it does not mention employee privacy concerns, which is an important aspect of Manager T's argument. Therefore, only the third answer choice could support Manager T's argument. Moreover, since Manager T argues that the documentation could compromise employee privacy, which Manager T feels is very important, a concluding sentence (such as the third answer choice) that the company should not require the specific documentation that it does would complete Manager T's argument.

The correct answer is *Our company should not require that form of documentation*.

120. Ethics board member: All actions that are permissible under the code of ethics are also legal in all of the jurisdictions in which our company operates. Furthermore, regardless of whether it has been determined if an action is legal, it is always permissible to ask the ethics board to review the action for conformity to the code of ethics.

Statements: Without exception, an action is permissible under the code of ethics ____1____ it is legal in all of the jurisdictions in which the company operates. Furthermore, one is permitted to ask the ethics board to review an action for conformity to the code of ethics ____2____ the legality of the action has been established.

Select for *1* and for *2* the two different options that complete the statements so that they most accurately paraphrase the ethics board member's assertions. Make only two selections, one in each column.

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1	2	
		if
		only if
		unless
		whether or not
		or

Answer Explanation:

Recognize

***1*:**

The ethics board member's first sentence states that for all actions, if that action is permissible under the code of ethics, then it is legal in all of the jurisdictions in which the company operates. Since "for all actions" means without exception (when applied to actions), and a statement of the form "if P, then Q" is logically equivalent to "P only if Q," it follows that the ethics board member's first sentence is logically equivalent to saying that without exception, an action is permissible under the code of ethics **only if** it is legal in all of the jurisdictions in which the company operates.

To see that each of the other options gives a less accurate paraphrase, let P and Q represent the assertions as suggested above. Thus, the ethics board member's first sentence is logically equivalent to "for all actions, if P, then Q." The selection **if** gives a statement logically equivalent to "for all actions, if Q, then P" (not a paraphrase), the selection **unless** gives a statement logically equivalent to "for all actions, if not-Q, then P" (not a paraphrase), the selection **whether or not** gives a statement logically equivalent to "for all actions, P if and only if Q" (not a paraphrase), and the selection **or** gives a statement logically equivalent to "for all actions, P or Q" (not a paraphrase).

The correct answer is *only if*.

***2*:**

Since "whether it has been determined" means the same as "whether or not it has been determined," the ethics board member's second sentence is logically equivalent to "whether or not R, then S," where R represents "it has been determined if an action is legal" and S represents "it is permissible to ask the ethics board to review the action for conformity to the code of ethics." This is logically equivalent to "S whether or not R," which corresponds to the selection **whether or not**.

To see that each of the other options gives a less accurate paraphrase, let R and S represent the assertions as suggested above. Thus, the ethics board member's second sentence is logically equivalent to "if R or not-R, then S," which is logically equivalent to "if R, then S, and if not-R, then S." The selection **if** gives a statement logically equivalent to "if R, then S" (doesn't include the provision that we have S when not-R holds), the selection **only if** gives a statement logically equivalent to "if S, then R" (doesn't allow S to be concluded), the selection **unless** gives a statement logically equivalent to "if not-R, then S" (doesn't include the provision that we have S when R holds), and the selection **or** gives a statement logically equivalent to "S or R" (doesn't include the provision that we have S when R holds).

The correct answer is *whether or not*.

121. A corporation uses a model of diminishing returns to make predictions about the expected returns on research investment. For this model, in order to produce an $x\%$ increase in annual profits in subsequent years, the corporation must invest $y\%$ of annual profits into research, where $y = 2x^2$.

Select two different numbers that are jointly compatible with the information provided and could be the values for x and for y. Make only two selections, one in each column.

x	y	

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x	y	
		1
		3
		5
		20
		50
		80

Answer Explanation:

Infer

The following table shows the value of $2x^2$ for each possible selection for x.

x	$2x^2$
1	2
3	18
5	**50**
20	800
50	5,000
80	12,800

The table shows that among the possible selections for x, only $x = 5$ gives a value of $2x^2$ that is equal to one of the possible selections for $2x^2$. Therefore, $x = 5$ and $y = 50$.

x:

The correct answer is *5*.

y:

The correct answer is *50*.

122. Each of 1,455 runners registered to run in an upcoming marathon answered "Yes" or "No" to two questions on the registration form: (1) Do you run at least 25 miles per week? and (2) Do you experience joint pain? Of those who answered "Yes" to the first question, 785 answered "Yes" to the second question, and 380 answered "No" to the second question. Of those who answered "No" to the first question, x runners answered "Yes" to the second question, and y runners answered "No" to the second question.

Select for x and for y values that are jointly consistent with the given information. Make only two selections, one in each column.

x	y	
		75

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		145
		215
		290
		455

Answer Explanation:

Infer

Since 1,455 runners were asked the questions and 785 + 380 = 1,165 runners answered "Yes" to the first question, 1,455 − 1,165 = 290 runners answered "No" to the first question. These 290 runners consist of the x runners whose answers to the two questions were "No" followed by "Yes" and the remaining y runners whose answers to the two questions were "No" followed by "No." Therefore, $x + y = 290$. Among the available numbers, only 0 and 290 have a sum of 290.

x:

$x = 290$ because 290 is the greater of the numbers 0 and 290.

The correct answer is *290*.

y:

$y = 0$ because 0 is the smaller of the numbers 0 and 290.

The correct answer is *0*.

123. A corporation plans to hire several new administrators and several new administrative assistants. It has a budget of exactly 100,000,000 yen (¥) per year allocated to spend on salaries for these new hires. The corporation wants to hire enough new administrators and administrative assistants to use all of this budget, and no more, each full year after the hiring. The annual salary will be exactly ¥10,000,000 for each administrator and exactly ¥7,500,000 for each administrative assistant.

Statement: Hiring exactly ___X___ new administrators and exactly ___Y___ new administrative assistants would meet the company's stated goal.

Select for X and for Y the options that complete the statement so that it is accurate based on the information provided. Make only two selections, one in each column.

X	Y	
		2
		3
		4
		5
		6
		7

Answer Explanation:

Strategize

The total annual salary in millions of yen for X new administrators and Y new administrative assistants is $10X + 7.5Y$. Therefore, the company's stated goal will be met if $10X + 7.5Y = 100$, or

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$10X + 7.5Y$. Therefore, the company's stated goal will be met if $10X + 7.5Y = 100$, or $7.5Y = 100 - 10X$. Since X is an integer, $100 - 10X$ is an even integer, and hence the integer Y must be a multiple of 4. The only multiple of 4 among the options is 4, so $Y = 4$. Finally, using $10X + 7.5(4) = 100$, we get $X = 7$.

Alternatively, for the equation $10X + 7.5Y = 100$, we can use a type of trial and error, but rather than consider all 6 × 6 = 36 possible choices for X and Y, it will be more efficient to substitute each of the 6 options into one of the unknowns, solve for the other unknown, and then see whether the other unknown is among the 6 options. This is done in the table below, where Y is the unknown the 6 options are substituted into. The reason for beginning with Y is because then the last step, namely obtaining X from $10X = 100 - 7.5Y$, will require only division by 10, which can be done mentally and quickly.

Y	$100 - 7.5Y$	X
2	85	8.5
3	77.5	7.75
4	70	7
5	62.5	6.25
6	55	5.5
7	47.5	4.75

x:

In the first method above, it was determined that $X = 7$. Alternatively, in the table above, the only value of X that appears among the options is 7.

The correct answer is *7*.

y:

In the first method above, it was determined that $Y = 4$. Alternatively, in the table above, the value of Y that corresponds to $X = 7$ is 4.

The correct answer is *4*.

Questions 124 to 141 — Difficulty: Medium

124. A small business has just invested in a new piece of equipment. The business's accountant estimates that the value of the equipment will decrease over time at a constant rate. Based on this assumption, the accountant properly estimates the value of the equipment 3 years afterward to be $\$X$, 6 years afterward to be $\$Y$, and 8 years afterward to be, for the first time, $0.

Select values for X and for Y that are jointly consistent with the given information. Make only two selections, one in each column.

X	Y	
		200
		300
		400
		500

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		600

Answer Explanation:

Evaluate

X:

The accountant estimates that the equipment's value will decrease at a constant rate to $0 over eight years. If that's true, the decrease will be 1/8 of the equipment's current value per year for eight years. Thus, after 3 years the equipment will be worth 5/8 of its current value, and after 6 years it will be worth 2/8 of its current value. So, the value $\$Y$ after 6 years will be 2/5 of the value $\$X$ after 3 years. Among the five answer options, only $X = 500$ and $Y = 200$ are such that Y is 2/5 of X. Thus, $X = 500$ and $Y = 200$.

The correct answer is *500*.

Y:

As explained above, $Y = 200$.

The correct answer is *200*.

125. Pavel made a list of 31 chores he needed to complete over the next 7 days. On the first 5 days, he completed 3, 5, 7, 2, and 6 of the listed chores, respectively. On the next 2 days, he completed the remaining 8 chores.

Select for *Least possible median* and for *Greatest possible median* values that are consistent with the information provided and give the least and greatest possible values, respectively, for the median of the numbers of the listed chores Pavel completed on each of the 7 days. Make only two selections, one in each column.

Least possible median	Greatest possible median	
		2
		3
		4
		5
		6

Answer Explanation:

Evaluate

***Least possible median*:**

Over the final 2 of the 7 days, Pavel finished 8 chores, so for some whole numbers x and y, he must have finished exactly x chores on one day and y chores on the other day such that $x + y = 8$ and $x \leq y$. To find the median number of chores he finished per day over all 7 days, rank the days in order of how many chores he finished on each. The median is the number of chores he finished on the day ranked 4th out of the 7. If $x = y = 4$, then the ranking is {2, 3, 4, 4, 5, 6, 7}, so the median is 4. Otherwise, $y \geq 5$ and $x \leq 3$. In that case, the four days on each of which he finished the most chores must be the days on which he finished y, 5, 6, and 7 chores, respectively. Since $y \geq 5$, the number of chores he finished on the day ranked 4th out of the 7 must then be 5, so 5 must be the median. Thus, we've shown that the median number of chores must be either 4 or 5. So 4 is the least possible value of the median.

The correct answer is *4*.

***Greatest possible median*:**

As explained above, the median number of chores must be either 4 or 5. So 5 is the greatest possible value of the median.

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The correct answer is *5*.

126. Literary critic: When a reader feels an emotion that is focused on the events and characters in a work of fiction, the reader is psychologically somewhat detached from that emotion. Lacking the immediacy of emotions about events in the reader's own life, emotions evoked by fiction are enjoyed as pure sensations independent of ____1____. Consequently, the reader can find pleasure even in sadness when it is focused on the events and characters in fictional works, because the work's beauty consists partly in its ability to evoke such ____2____.

Select options to fill in the two blanks above to complete the literary critic's statements most coherently. Make only two selections, one in each column.

1	2	
		beauty
		psychological detachment
		typically unpleasant emotions
		fictional events
		real events

Answer Explanation:

Evaluate

***1*:**

We need to determine the word or phrase that best fills blank 1 to complete the sentence. To correctly complete the sentence, we must determine what emotions, evoked by fiction and enjoyed as pure sensations, are independent of. At the beginning of the second sentence, we are told that emotions evoked by fiction lack "the immediacy of emotions about events in the reader's own life." It seems reasonable to say here that the emotions evoked by fiction are enjoyed as pure sensations independent of *real events*.

The correct answer is *real events*.

***2*:**

We need to determine the word or phrase that best fills blank 2 to complete the sentence. Once the blank is appropriately filled, the sentence is supposed to support—along with the sentence completed by the answer that fills blank 1—the claim that a reader of fiction can find pleasure in sadness when the sadness is focused on events and characters in fictional works. Which of the possible responses would help create a sentence that would support that claim? Sadness is typically an unpleasant emotion. So, if a work of fiction's beauty consists in its ability to evoke typically unpleasant emotions such as sadness, but the sadness can be evoked by fiction without the unpleasantness, then a reader can find pleasure even in sadness, as long as the sadness is evoked independent of real events (i.e., as when it is focused on the events and characters in fictional works).

The correct answer is *typically unpleasant emotion*.

127. The equation $6 \times 5 - 4 \div 2 - 3 + 7)^0 - 1)^2 = 6$ is missing two opening parentheses. It can be made true if opening parentheses are placed immediately before each of two of the numbers.

Assume that standard order-of-operations principles are followed, whereby multiplication and division are performed before addition and subtraction, unless parentheses indicate otherwise. Select in the table a number before which the first opening parenthesis could be placed and a number before which the second opening parenthesis could be placed, such that the equation would express a true statement. Make only two selections, one in each column.

First parenthesis immediately before	Second parenthesis immediately before	
		2

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		2
		3
		4
		5
		6

Answer Explanation:

Apply

Try all possible placements of the two opening parentheses:

Before 5 and 4	$6 \times (5 - (4 \div 2 - 3 + 7)^0 - 1)^2 = 54$
Before 5 and 2	$6 \times (5 - 4 \div (2 - 3 + 7)^0 - 1)^2 = 0$
Before 5 and 3	$6 \times (5 - 4 \div 2 - (3 + 7)^0 - 1)^2 = 6$
Before 5 and 7	$6 \times (5 - 4 \div 2 - 3 + (7)^0 - 1)^2 = 0$
Before 5 and 1	$6 \times (5 - 4 \div 2 - 3 + 7)^0 - (1)^2 = 5$
Before 4 and 2	$6 \times 5 - (4 \div (2 - 3 + 7)^0 - 1)^2 = 21$
Before 4 and 3	$6 \times 5 - (4 \div 2 - (3 + 7)^0 - 1)^2 = 30$
Before 4 and 7	$6 \times 5 - (4 \div 2 - 3 + (7)^0 - 1)^2 = 29$
Before 4 and 1	$6 \times 5 - (4 \div 2 - 3 + 7)^0 - (1)^2 = 28$
Before 2 and 3	$6 \times 5 - 4 \div (2 - (3 + 7)^0 - 1)^2 =$ undefined division by 0
Before 2 and 7	$6 \times 5 - 4 \div (2 - 3 + (7)^0 - 1)^2 = 26$
Before 2 and 1	$6 \times 5 - 4 \div (2 - 3 + 7)^0 - (1)^2 = 25$
Before 3 and 7	$6 \times 5 - 4 \div 2 - (3 + (7)^0 - 1)^2 = 10$
Before 3 and 1	$6 \times 5 - 4 \div 2 - (3 + 7)^0 - (1)^2 = 26$
Before 7 and 1	$6 \times 5 - 4 \div 2 - 3 + (7)^0 - (1)^2 = 25$

***First parenthesis immediately before*:**

The third row of the table shows an expression with value 6. The first opening parenthesis is before 5.

The correct answer is *5*.

***Second parenthesis immediately before*:**

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Second parenthesis immediately before.

The third row of the table shows an expression with value 6. The second opening parenthesis is before 3.

The correct answer is *3*.

128. Historian: New York City became an influential art center in the twentieth century, as seen, for example, in the influential 1913 Armory Show and the growth of museums. This was due primarily to a combination of New York's large and varied population, its basic resources, and its access to European art trends. The city could never have become an influential art center without the era's increased local wealth, which allowed more people to buy art. However, the changes were driven principally by the diversity of the city's population, including intellectuals and artists from Europe, and a culture that prized energy and creativity, which stimulated the artistic endeavors of many people.

Select for *Precondition* the statement that the passage most strongly suggests describes a precondition for New York becoming an influential art center. And select for *Strengthener* the statement that would, if true, most clearly strengthen the support for the historian's explanation of why the city became an influential art center. Make only two selections, one in each column.

Precondition	**Strengthener**	
		New York showed significant economic development leading up to the early twentieth century.

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